

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (E&E) (Credit System) Degree Examinations

May - June 2023

20EE601 – POWER SYSTEM ANALYSIS AND STABILITY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I

Marks BT* CO* PO*

1. a) What is per unit quantity? Mention the advantages of per unit quantities.
- b) Figure Q. 1(b) shows the reactance diagram of a radial transmission system. The ratings and reactances of various components are as shown therein. A load of 60 MW at 0.9 p.f. lagging is tapped from the 66 kV substation which is to be maintained at 60 kV. Calculate the terminal voltage of the machine. Represent the transmission line and transformer by series reactance only. Take base values as 100 MVA and 220 kV on the transmission line.

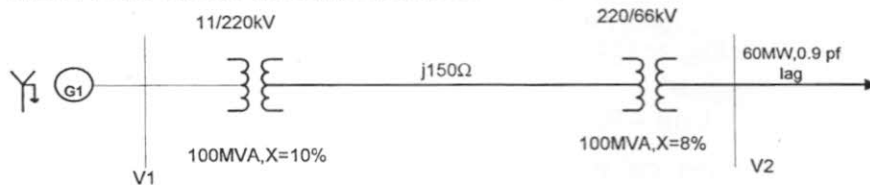


Figure Q1b.

2. a) Show that the per unit reactance of a two-winding transformer will remain the same whether it is referred to primary or secondary.
- b) A 100 MVA, 33 kV, 3-phase generator has a sub-transient reactance of 15%. The generator is connected to three motors through a transmission line and transformers as shown in Fig. Q2(b). The motors are rated for 30 MVA, 20 MVA and 50 MVA at 30 kV with 20% subtransient reactance each. The transformers are rated 100 MVA, 32 kV Delta / 100 kV Star with leakage reactance of 8%. The line has a reactance of 50 Ω. Selecting the generator ratings as the base in the generator circuit, draw the per unit reactance diagram.



Figure Q2b

- c) Describe the assumptions made for obtaining the reactance diagram.

Unit – II

3. a) With a neat oscillogram, describe subtransient, transient and steady-state symmetrical fault currents of a synchronous generator. 10 L2 2 1
- b) A synchronous generator rated 500 kVA, 440 V, 0.1 p.u. subtransient reactance is supplying a passive load of 400 kW at 0.8 p.f. lag. Calculate the initial symmetrical current (rms) for a fault at the generator terminals. Take 500 kVA and 440 V as the base values for the generator. 10 L3 2 2
4. a) Define short-circuit MVA of a system. Describe its significance. 8 L2 2 2
- b) A three-phase, 5 MVA, 6.6 kV alternator with a reactance of 8% is connected through a transformer to a feeder of series impedance of $(0.12 + j0.48)$ ohm/phase per km as shown in Figure Q4(b). The transformer is rated at 3MVA, 6.6 kV/33 kV and has a series reactance of 5%. Determine the fault current supplied by the generator operating under no load with a voltage of 6.9 kV, when a three-phase symmetrical fault occurs at a point 15 km along the feeder. Choose the base values for the generator side as 5 MVA, 6.6 kV.

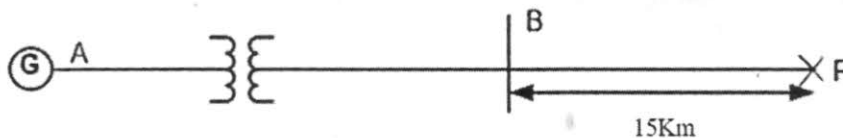


Figure Q4b

12 L3 2 2

Unit – III

5. a) Show that in case of a symmetrical three-phase load, current of a particular sequence produces voltage drop of the same sequence only. 8 L2 3 1
- b) Draw the zero sequence diagrams for the following configurations of three-phase transformers:
- (i) Ungrounded star / ungrounded star
 - (ii) Grounded star/ ungrounded star
 - (iii) Grounded star / grounded star
 - (iv) Ungrounded star / delta
 - (v) Grounded star / delta
 - (vi) Delta / delta.
- Take the grounding impedance as zero Ω . 12 L2 3 2
6. a) Prove that a balanced set of three-phase voltages has only positive sequence components of voltages. 5 L2 3 1
- b) The line voltages at the terminals of a balanced load consisting of three 20-ohm star-connected resistors are: 200 V at angle of 0 degree, 100 V at angle of 255.5 degree and 200 V at an angle of 151 degree. Find the power supplied to the load using the symmetrical components. Assume that the neutral of the load is isolated. 10 L3 3 2

- c) Write a brief note on the sequence impedances of a three-phase alternator.

5 L1 3 1

Unit – IV

7. a) Derive an expression for fault current for line-line fault on an unloaded generator.
- b) A salient-pole generator without dampers is rated 20 MVA, 13.8 kV and has a subtransient reactance of 0.25 p.u. The negative and zero sequence reactances are 0.35 p.u. and 0.1 p.u. respectively. The neutral of the generator is solidly grounded. Calculate the fault current in Amperes when a single line-to-ground fault occurs at the terminals of the generator. Assume pre-fault current is zero. Take generator ratings as the base values.

10 L2 4 1

10 L3 4 2

8. a) Derive an expression for fault current for a single line-ground fault on an unloaded generator.
- b) The following data may be assumed for the network shown in Fig. Q8(b).
 Generator: 50 MVA, 11 kV, $X_1 = 80\%$, $X_2 = 50\%$, $X_0 = 20\%$.
 Transformer: 40 MVA, 11/110 kV, $X_1 = X_2 = X_0 = 6\%$.
 The percentage reactances are given on the base of ratings of each device. If a double line-ground fault occurs at F, find the conductor current in Amperes at the point of fault. Take 50 MVA and 11 kV as the base values for the generator circuit.

8 L2 4 1

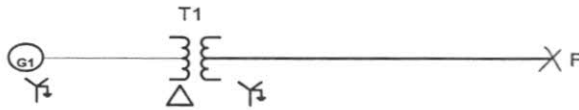


Figure Q8b.

12 L3 4 2

Unit – V

9. a) Derive power-angle equation for a salient pole synchronous machine.
- b) A salient-pole synchronous machine having $X_d = 0.6$ p.u. and $X_q = 0.4$ p.u. per phase is operated from an infinite bus of voltage 1 p.u. If the excitation voltage is 1.1 p.u., find the steady-state stability limit in p.u. and the rotor angle at which this occurs.
10. a) Write a note on equal-area criterion.
- b) A 6-pole, 50 Hz turbogenerator of capacity 80 MW working at 0.8 p.f. has an inertia of 10 MJ/MVA.
 (i) Calculate the energy stored in the rotor at synchronous speed.
 (ii) Find the rotor acceleration if the mechanical input is suddenly raised to 75 MW for an electrical load of 60 MW.
 (iii) Supposing the above acceleration is maintained for a duration of 6 cycles, calculate the change in torque angle.

10 L2 5 1

10 L3 5 2

10 L1 5 1

10 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
